



ROOT CAUSE UNLOCKED · CONDITION KIT

The Migraine Root Cause *Kit*

Which tests actually matter, what the numbers mean, and exactly what to ask your doctor for

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The Migraine Root Cause Kit

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Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Go to an emergency department now, do not finish this page, for any of these:

- A headache that reached maximum intensity within a minute, often described as the worst of your life
- Headache with fever and a stiff neck
- Headache with new weakness, numbness, trouble speaking, vision loss, or confusion
- Headache after a head injury
- A first severe headache after age 50
- Headache that is clearly worse when you lie flat or when you stand up
- Headache in pregnancy or right after delivery, or with a history of cancer or a suppressed immune system

That list is the practical version of the SNNOOP10 red flag list, which was tested prospectively in an emergency department against neurologist diagnosis to see how well it catches high-risk headaches (PMID 36003002). None of those features are migraine behaving normally, and none of them are a reason to run a nutrient panel first.

The honest starting point

Migraine is a clinical diagnosis. There is no blood test for it.

It is diagnosed by pattern: attack duration, the character of the pain, what comes with it, and how it behaves over time, against published criteria (PMID 29368949). Anyone selling you a lab panel that diagnoses migraine is selling you something.

So blood work does two jobs here: it finds the things that mimic or worsen migraine and are correctable (iron deficiency, thyroid disease, celiac disease, poor glycemic control), and it finds the modifiable nutrient inputs that change how often attacks come.

That is a smaller claim than most functional medicine marketing makes, and it is still worth doing, because the correctable things on that list are common and cheap to fix. It is also only part of the workup, because three of the five root causes below are not blood tests at all.

The five keys, and what tests each one

Unlock Your Migraines organizes migraine around five root causes. Testing exists to find out which of them are yours, so this guide is organized the same way.

Key	What it means	How you find it
1. Food	Delayed food reactivity, meal timing, and glycemic swings feeding attack frequency	Diary, food-specific IgG panel, structured elimination and reintroduction
2. Hormones	Attacks tracking the menstrual cycle, perimenopause, or hormonal contraception	Cycle-mapped diary. Not a blood panel
3. Chronic infection	An infectious or inflammatory load acting as a background driver	H. pylori testing where gut symptoms coexist
4. Structure	Neck, upper cervical, jaw, and postural contribution	Physical examination. Not a blood test
5. Nutrients	Magnesium, riboflavin, CoQ10, iron, vitamin D, and the methylation pathway	Blood work, which is most of this guide

Two of those five have no blood test at all, and one of them, structure, is the one most often left out of a migraine workup entirely. That does not make them less real. It makes them the ones you have to look for on purpose.

The highest-yield question in this guide is not a lab test

Count your acute medication days. How many days per month do you take something to stop a headache, including over-the-counter combination products, and including the caffeine-containing ones.

If that number is regularly 10 or more, medication overuse headache is likely part of your picture, and no supplement, diet, or lab panel will outperform addressing it. A network meta-analysis of management strategies found that abrupt withdrawal on its own was not significantly effective, with a mean difference of -2.77 monthly headache days and a confidence interval crossing zero, while withdrawal combined with a preventive strategy was effective, and combination approaches including anti-CGRP therapies and nerve blocks ranked highest (PMID 40011869).

Read that carefully, because it cuts against the usual advice. “Just stop taking them” is the version that did not work. Stopping with a preventive plan in place is the version that did. That is a conversation with a prescriber, and it is the single most valuable thing many readers of this guide could do.

How to read this guide

Where I give a functional range, it is a clinical target used at Root Health, not a validated diagnostic threshold, and it sits inside the standard laboratory range rather than replacing it. Being outside a functional range is a reason to look closer, not a diagnosis.

This guide runs in the order of the five keys: the nutrient and mimic blood work first, because that is what a lab draw answers, then hormones, chronic infection, and structure, which are assessed differently. The food key has its own section, because in migraine personalized elimination has better evidence behind it than in almost any other condition I write about.

Tier 1: contributors worth finding, because they are correctable

Iron and ferritin

Why it is first. Migraine is roughly three times more common in women than men, iron deficiency is common in menstruating women, and the two overlap constantly. A literature review of iron deficiency anemia and migraine reports a higher prevalence of iron deficiency anemia among people with migraine, particularly women, with several studies describing an inverse relationship between serum ferritin and migraine severity, while noting that iron supplementation remains a promising rather than established therapy and that iron overload is a real risk in people who do not need it (PMID 39429346).

What to ask for. Ferritin, serum iron, TIBC, transferrin saturation, and a CBC, all together. Ferritin alone is not enough, because it rises with inflammation and can look reassuring in someone who is genuinely depleted.

How to read it. A ferritin in the teens or twenties with heavy periods and fatigue is worth acting on even when the hemoglobin is still normal, because iron stores empty before the blood count changes. Correcting it is a medical decision with a real dose and a retest, not a guess.

Vitamin D

Why it is here. A dose-response meta-analysis of 22 randomized trials of dietary supplements for migraine prophylaxis found that vitamin D reduced migraine frequency, with a mean difference of -1.69 attacks, and reduced monthly migraine days by -2.41 (PMID 39404918). Two randomized, triple-blinded, placebo-controlled trials of vitamin D given together with a probiotic in adults with migraine reported greater reductions in headache frequency and severity, and in a composite migraine index, than placebo (PMID 39394141, PMID 40360751).

How to read it. Those trials are the reason to test rather than the reason to megadose. Get the number, correct a real shortfall with your clinician, and retest.

Magnesium, and an honest word about measuring it

Why it matters. Magnesium is the best-supported supplement in migraine prevention, which is covered in the supplement guide. Testing it is harder than supplementing it.

The problem with the test. Most magnesium in the body is inside cells and bone, so a serum level is a poor window on total status. Red blood cell magnesium is generally the better of the two available options. A study measuring both plasma and erythrocyte magnesium in 25 people with migraine against 25 controls found both to be significantly lower in the migraine group (PMID 36354242). That is a small study, and it is a reasonable justification for looking rather than proof of a threshold.

What I would do. Ask for RBC magnesium if you want the number. Understand that a normal result does not rule out a magnesium-responsive migraine pattern, because the trials that showed benefit did not select patients by their magnesium level. In practice, the trial of supplementation is more informative than the test.

Homocysteine, B12, and folate

Why this cluster. There is a real line of randomized evidence here, and it is specific to migraine with aura. A randomized, double-blind, placebo-controlled trial gave 52 patients with migraine with aura daily folic acid 2 mg, vitamin B6 25 mg, and vitamin B12 400 micrograms for six months. Homocysteine fell by 39%, about 4 micromol/L, significantly more than placebo, and headache frequency and pain severity were reduced in the treated group while the placebo group did not change. The treatment effect on both homocysteine and migraine disability was associated with MTHFR C677T genotype, with carriers of the C allele responding better than TT genotypes (PMID 19384265).

A larger 206-patient randomized trial of the same B vitamin combination found that both MTHFR C677T and MTRR A66G variants independently influenced the response, with C allele carriers and A allele carriers respectively showing greater reductions in homocysteine, pain severity, and disability (PMID 22926161). Separately, in a cohort of 141 women with migraine with aura, higher dietary folate intake was inversely associated with migraine frequency (PMID 25598270).

What to ask for. Homocysteine, B12, and RBC folate. MTHFR genotyping is optional and is genuinely informative here in a way it is not in most contexts, because in these trials genotype predicted who responded. Note the direction of that finding, which surprises people: the C allele carriers, not the TT homozygotes, showed the larger response.

Thyroid and blood sugar

Thyroid. Both underactive and overactive thyroid produce headache along with fatigue, and thyroid disease is common enough in the same demographic that a TSH with free T4 is a reasonable part of any first migraine panel.

Blood sugar. Skipped meals and glycemic swings are among the most consistently reported migraine triggers, and a review of fasting and headache notes that genetic studies have found a link between glycemic processing and migraine occurrence (PMID 40117066). Fasting glucose and HbA1c are cheap, and they occasionally reframe the “I get a headache if I do not eat” pattern as something measurable.

Tier 2: the mimics worth excluding once

Celiac disease

This one is under-tested in headache and it is the clearest rule-out on the list.

A systematic review and meta-analysis of 40 articles on headache and celiac disease found an increased prevalence of celiac disease among people with idiopathic headache and vice versa, reported the prevalence of celiac disease in children with idiopathic headache at 2.4%, described the headaches as most often migraine-like, and found that a gluten-free diet led to total resolution of headache in up to 75% of patients in that context (PMID 30301194). A review of the question concludes that the evidence supports screening for celiac disease during the initial diagnosis of recurrent headache in both adults and children (PMID 38731144). An online survey of 866 people with biopsy-confirmed celiac disease and headache found that after starting a gluten-free diet, headache frequency and intensity improved significantly more in those with migraine than with tension-type headache, and that compliant individuals showed a 48% improvement in headache frequency (PMID 30893404).

The rule that gets broken constantly: celiac serology must be drawn while you are still eating gluten.

Remove gluten first and the test becomes uninterpretable, and you will never get a clean answer without a gluten challenge you will not enjoy. Test first, then experiment.

Blood pressure and sleep

Neither is a blood test, and both belong in this section.

Get an accurate blood pressure. And if you snore, wake unrefreshed, or wake with headache, ask about a sleep apnea assessment. Morning headache is a classic presentation, and treating the apnea treats the headache in a way no supplement will.

The food question, where migraine has better evidence than most conditions

Food-specific IgG panels are controversial in general. In migraine specifically, they are one of the few places where the controversy has actually been tested with randomized trials, and the results are worth knowing before you form an opinion.

The strongest study. A randomized, double-blind, crossover, headache-diary-based trial enrolled 30 patients with migraine without aura. After a six-week baseline, IgG antibodies against 266 food antigens were measured, and patients were randomized to six weeks of a diet either excluding or including their own IgG-positive foods, then crossed to the opposite diet after a two-week washout. Participants averaged 24 foods with abnormally high titre. During the elimination period, headache days fell from 10.5 to 7.5 and migraine attacks fell from 9.0 to 6.2 compared with baseline, both statistically significant (PMID 20647174).

The trial that tempers it. A randomized controlled trial of 167 community volunteers with migraine-like headaches compared a true elimination diet based on ELISA IgG testing against a sham diet sheet for 12 weeks. The true diet significantly reduced headaches at the four-week assessment, but by 12 weeks the difference was no longer statistically significant, and disability and impact on daily life did not differ between groups (PMID 21835022).

The summary from the headache literature. A narrative review of diet and headache states that two of three randomized controlled trials reported that an elimination diet of IgG positive foods significantly decreased the frequency of headache or migraine during treatment compared with baseline, and concludes that elimination diets can prevent headaches in subgroups of people with headache disorders (PMID 27699780).

How I use this. A panel gives you a personal short list instead of a generic trigger list, which is the practical problem it solves. A blind elimination diet in someone with 266 candidate foods is not a project most people finish. The panel narrows it, and the reintroduction confirms it. The Migraine Diet Guide walks through exactly how to run that, including the reintroduction step that turns a list into an answer.

Two limits worth carrying. The 12-week result above says that adherence and effect can fade, so plan the reintroduction rather than assuming a permanent diet. And an IgG panel is not an allergy test: it cannot rule an IgE food allergy in or out, and a low IgG value is never a reason to eat something that has caused a serious allergic reaction.

Key 2: hormones, which a blood panel will not show you

If your attacks cluster in the two days before your period and the first three days of it, that pattern is the finding, and no hormone panel will confirm it better than a calendar.

What is happening. Menstrual migraine affects roughly 6% of reproductive-age women, and the leading explanation is estrogen withdrawal: the premenstrual fall in estrogen acts as the trigger. A 2023 review revisiting that hypothesis notes it is widely accepted while critically appraising how strong the supporting evidence actually is (PMID 37730536). It is recognized as a distinct disorder that needs greater recognition than it usually gets (PMID 33600767), and the broader hormonal picture involves interactions between estrogen, oxytocin, and CGRP (PMID 34545218).

Why measuring hormones usually disappoints. A single serum estradiol tells you about one moment in a cycle. It is the *change* that matters, and the practical way to capture change is to map attacks against cycle days for three months.

What to bring instead of a lab result. Three cycles of dates: period start, attack days, severity. That single sheet of paper is what makes a targeted plan possible, and it is what a clinician needs before considering cycle-timed prevention.

The safety line that belongs here. Migraine **with aura** plus combined hormonal contraception is a specific stroke-risk conversation with a prescriber. If you have aura and take a combined pill, raise it at your next appointment rather than stopping anything on your own.

Key 3: chronic infection, stated with the honesty it deserves

The ebook treats infectious and inflammatory load as a background driver. Here is where the testable evidence actually sits.

H. pylori. A systematic review and meta-analysis examined the association between H. pylori infection and migraine, and the effects of the infection and its eradication on migraine characteristics, noting that two decades of studies have produced different and sometimes contradictory results (PMID 37529895). That is a genuine association signal in a contested literature, not a settled cause.

How I use it. I do not screen every migraine patient for H. pylori. I test when there are upper gut symptoms alongside the headaches, because then the test is justified on its own terms and the migraine question becomes a bonus. The lab guide for reflux covers how to test properly, including the washout from acid suppression.

The wider gut-brain link. A comprehensive review of the gut-brain axis in migraine describes the bidirectional signalling involved (PMID 32054443), and a 2026 review examines microbiome-targeted approaches specifically (PMID 41829891). Read those as a mechanism worth understanding rather than a test to order. There is no microbiome panel that currently tells you what to do about your migraine.

Key 4: structure, which is a physical examination and not a lab

This is the key most often missing from a migraine workup, and it is the one I am trained to assess.

Neck involvement is common and measurable. Neck pain is prevalent in migraine, and musculoskeletal dysfunctions are prevalent enough that they are a legitimate physiotherapy target rather than an incidental finding (PMID 37331926). A cross-sectional study comparing women with episodic migraine, chronic migraine, and controls examined upper cervical mobility, symptomatic upper cervical joints, cervicocephalic kinesthesia, and head and neck posture, and found differences in musculoskeletal impairments of the cervical spine between the groups (PMID 28118694). A review of physical therapy in migraine describes both musculoskeletal and balance dysfunctions and why they matter in practice (PMID 31813696).

What this does not mean. It does not mean your migraine is “really” a neck problem, and it does not mean an adjustment cures migraine. Neck involvement is common in migraine, and it is often a consequence as much as a contributor.

What it does mean. A structural assessment is a reasonable part of a migraine workup, particularly if you have neck pain, jaw clicking or clenching, a desk posture you hold for hours, or attacks that reliably follow physical positions. Ask for an assessment of upper cervical mobility, jaw function, and head and neck posture.

Key 5: nutrients, and the stress axis that sits underneath everything

The nutrient testing is above. The piece the first version of this guide left out is the stress axis, and in migraine it is not a soft factor.

Stress moves through the migraine cycle in a measurable pattern. A longitudinal cohort study using daily prospective diary data described patterns of perceived stress across the stages of the migraine cycle, within and between individuals (PMID 32918830). Stress is not simply a trigger sitting outside the attack; it varies with the phases of the attack itself.

And the intervention evidence is good, which is the part that matters. A systematic review and meta-analysis of behavioral interventions for migraine prevention synthesized benefits and harms in children and adults (PMID 39968795), and a separate systematic review and meta-analysis assessed biofeedback specifically (PMID 40064357). These are real preventive interventions with real evidence, not relaxation advice tacked onto the end of a plan.

How to assess it. Not with a cortisol panel. With the same diary you are already keeping: add a daily stress rating out of ten and a sleep quality rating out of ten next to your headache score. Three months of that shows the relationship far better than any single measurement of a stress hormone.

Functional ranges for the markers in this guide

Clinical targets used at Root Health, not validated diagnostic thresholds, narrower than the standard laboratory range on purpose.

Marker	Units	Functional target	Approximate standard range
Ferritin	ng/mL	50 to 150	25 to 300
Serum iron	ug/dL	65 to 110	40 to 150
TIBC	ug/dL	290 to 370	250 to 400
Transferrin saturation	%	20 to 40	15 to 45
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Magnesium (RBC)	mg/dL	5.2 to 6.8	4.0 to 7.2
Magnesium (serum)	mg/dL	2.0 to 2.5	1.5 to 2.7
Homocysteine	umol/L	under 7.0	under 10.0
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000
Vitamin B6 (P5P)	nmol/L	30 to 100	20 to 120
Glucose, fasting	mg/dL	75 to 86	70 to 99
HbA1c	%	4.6 to 5.3	4.0 to 5.6
hs-CRP	mg/L	under 0.5	under 1.0
White blood cells	K/uL	5.0 to 7.5	3.5 to 10.5
Albumin	g/dL	4.2 to 4.8	3.5 to 5.0

Reading your results as a pattern

Pattern one: the depleted migraineur. Low ferritin, low vitamin D, heavy periods, and attacks clustering premenstrually. Correct the iron and the vitamin D, and expect the benefit to show up over two to three months rather than two weeks.

Pattern two: migraine with aura and a raised homocysteine. This is the group with the most specific trial evidence in this whole guide (PMID 19384265, PMID 22926161). It has a defined intervention and a genotype that predicts response.

Pattern three: everything is normal. Common, and not a dead end. It means the levers are behavioral and pharmacologic rather than nutritional: acute medication days, sleep regularity, meal timing, caffeine consistency, and whether you are on a preventive that suits you. Several of those outperform anything a supplement will do.

Pattern four: the picture that does not fit migraine. Headaches that changed character, that are strictly one-sided in the same place every time, that wake you from sleep, or that came with any red flag at the top of this guide. Those go to a physician, not to a lab panel.

The conversation with your doctor

A request list you can bring

Reasonable in most primary care settings:

- CBC with differential
- Ferritin, serum iron, TIBC, transferrin saturation
- 25-hydroxy vitamin D
- TSH with free T4
- Homocysteine, vitamin B12, RBC folate
- Fasting glucose and HbA1c
- hs-CRP
- Celiac serology, drawn while still eating gluten
- A sleep apnea assessment if you snore or wake unrefreshed

Not laboratory tests, and two of the five keys depend on them:

- **A structural assessment:** upper cervical mobility, jaw function, and head and neck posture (Key 4)
- **Three cycles of a cycle-mapped headache diary**, if you menstruate (Key 2)
- **A daily stress and sleep rating** alongside your headache score

Usually through a functional medicine practice, often out of pocket:

- RBC magnesium
- Vitamin B6 as P5P
- MTHFR C677T genotype, if you have migraine with aura and a raised homocysteine
- A food-specific IgG panel, as the starting map for a structured elimination and reintroduction

The sentence that gets you taken seriously

Bring a diary, not a description. Attack dates, duration, severity, and acute medication days for one month is more persuasive than any adjective, and it is the same data any preventive decision will be based on. Lead with “I am having X headache days and taking something on Y of them.”

What these labs cannot tell you

- **Whether you have migraine.** That is a clinical diagnosis (PMID 29368949).
- **Which foods trigger your attacks.** A panel narrows candidates. Only elimination and reintroduction answers it.
- **Whether a supplement is working.** Nothing here is a progress measure. Monthly migraine days is the progress measure.
- **Why you have migraine at all.** Migraine is substantially genetic. Nothing in a normal panel explains it, and a normal panel is not a verdict on whether your pain is real.

When to seek clinical review

Emergency, repeating from the top because it matters most: thunderclap onset, fever with stiff neck, new neurological symptoms, head injury, first severe headache after 50, headache clearly triggered by lying flat or standing, headache in pregnancy or postpartum, or headache with cancer or immune suppression (PMID 36003002).

Soon:

- 10 or more acute medication days per month
- 4 or more migraine days per month, which is generally the threshold where preventive treatment is discussed
- Attacks that have changed in character, frequency, or severity
- Aura that lasts more than an hour, or aura for the first time after age 50
- Migraine with aura in someone taking combined hormonal contraception, which is a specific stroke-risk conversation to have with a prescriber

Your next step

Testing tells you what you are working with. The companion guides do the rest: **The Migraine Diet Guide** covers meal timing, caffeine, and how to run the elimination and reintroduction properly, and **The Migraine Supplement Guide** covers what has been studied for prevention, at what dose, and for how long before you judge it.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=migraines-labs

Consultations are available to people located in North Carolina, where Dr. Seay is licensed.

If you are outside North Carolina, start the diary. In migraine it is not a soft recommendation, it is the instrument every treatment decision is measured against:

Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=migraines-labs

References

All references were verified against live PubMed records on 2026-08-08, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Migraines-Kit-Citation-Ledger.md`.

1. Garcia-Azorin D, Abelaira-Freire J, Gonzalez-Garcia N, et al. Sensitivity of the SNNOOP10 list in the high-risk secondary headache detection. *Cephalalgia*. 2022;42(14):1521-1531. PMID 36003002
2. Headache Classification Committee of the International Headache Society. The International Classification of Headache Disorders, 3rd edition. *Cephalalgia*. 2018;38(1):1-211. PMID 29368949
3. Koonalintip P, Yamutai S, Setthawatcharawanich S, et al. Network meta-analysis comparing efficacy of different strategies on medication-overuse headache. *J Headache Pain*. 2025;26(1):43. PMID 40011869
4. Al-Qassab ZM, Ahmed O, Kannan V, et al. Iron Deficiency Anemia and Migraine: A Literature Review of the Prevalence, Pathophysiology, and Therapeutic Potential. *Cureus*. 2024;16(9):e69652. PMID 39429346
5. Talandashti MK, Shahinfar H, Delgarm P, et al. Effects of selected dietary supplements on migraine prophylaxis: A systematic review and dose-response meta-analysis of randomized controlled trials. *Neurol Sci*. 2025;46(2):651-670. PMID 39404918
6. Tirani SA, Khorvash F, Saneei P, et al. Effects of probiotic and vitamin D co-supplementation on clinical symptoms, mental health, and inflammation in adult patients with migraine headache: a randomized, triple-blinded, placebo-controlled trial. *BMC Med*. 2024;22(1):457. PMID 39394141
7. Tirani SA, Saneei P, Khorvash F, et al. Effects of probiotic and vitamin D co-supplementation on migraine index, quality of life, and oxidative stress in adults with migraine headache: a randomized triple-blinded clinical trial. *Eur J Nutr*. 2025;64(4):179. PMID 40360751
8. Hoshino MT, Bochio MG, Spagnol Bonache J, et al. Plasma and erythrocyte magnesium levels: from validation of the method to analysis in volunteers diagnosed to migraine. *Magnes Res*. 2022;35(2):51-61. PMID 36354242
9. Lea R, Colson N, Quinlan S, et al. The effects of vitamin supplementation and MTHFR (C677T) genotype on homocysteine-lowering and migraine disability. *Pharmacogenet Genomics*. 2009;19(6):422-428. PMID 19384265
10. Menon S, Lea RA, Roy B, et al. Genotypes of the MTHFR C677T and MTRR A66G genes act independently to reduce migraine disability in response to vitamin supplementation. *Pharmacogenet Genomics*. 2012;22(10):741-749. PMID 22926161
11. Menon S, Lea RA, Ingle S, et al. Effects of dietary folate intake on migraine disability and frequency. *Headache*. 2015;55(2):301-309. PMID 25598270
12. Ebbert PT, Natbony LR. Fasting and Headache. *Curr Pain Headache Rep*. 2025;29(1):67. PMID 40117066
13. Zis P, Julian T, Hadjivassiliou M. Headache Associated with Coeliac Disease: A Systematic Review and Meta-Analysis. *Nutrients*. 2018;10(10):1445. PMID 30301194
14. Perilli L, Carbone S, Novelletto LF, et al. Should We Rule out Celiac Disease in Recurrent Headache Disorders? A Review of the Literature. *J Clin Med*. 2024;13(9):2551. PMID 38731144
15. Ameghino L, Farez MF, Wilken M, et al. Headache in Patients with Celiac Disease and Its Response to the Gluten-Free Diet. *J Oral Facial Pain Headache*. 2019;33(3):294-300. PMID 30893404
16. Alpaya K, Ertas M, Orhan EK, et al. Diet restriction in migraine, based on IgG against foods: a clinical double-blind, randomised, cross-over trial. *Cephalalgia*. 2010;30(7):829-837. PMID 20647174

17. Mitchell N, Hewitt CE, Jayakody S, et al. Randomised controlled trial of food elimination diet based on IgG antibodies for the prevention of migraine like headaches. *Nutr J.* 2011;10:85. PMID 21835022
18. Martin VT, Vij B. Diet and Headache: Part 1. *Headache.* 2016;56(9):1543-1552. PMID 27699780
19. Raffaelli B, Do TP, Chaudhry BA, et al. Menstrual migraine is caused by estrogen withdrawal: revisiting the evidence. *J Headache Pain.* 2023;24(1):131. PMID 37730536
20. Ornello R, Caponnetto V, Ratti S, et al. Menstrual migraine: a distinct disorder needing greater recognition. *Lancet Neurol.* 2021. PMID 33600767
21. Krause DN, Warfvinge K, Haanes KA, et al. Hormonal influences in migraine, interactions of oestrogen, oxytocin and CGRP. *Nat Rev Neurol.* 2021. PMID 34545218
22. Bawand R, Ghiasian M, Samadyan M, et al. Association of Helicobacter pylori with migraine headaches and the effects of this infection and its eradication on the migraine characteristics in adults: A comprehensive systematic review and meta-analysis. *Helicobacter.* 2023;28(5):e13010. PMID 37529895
23. Arzani M, Jahromi SR, Ghorbani Z, et al. Gut-brain Axis and migraine headache: a comprehensive review. *J Headache Pain.* 2020. PMID 32054443
24. Migraine and the Gut-Brain Axis, The Role of Microbiome-Targeted Biotics. *Nutrients.* 2026. PMID 41829891
25. Luedtke K, Carvalho G, Szikszay T. Musculoskeletal dysfunctions and physiotherapy treatment strategies in patients with migraine. *Musculoskelet Sci Pract.* 2023;66:102805. PMID 37331926
26. Ferracini GN, Florencio LL, Dach F, et al. Musculoskeletal disorders of the upper cervical spine in women with episodic or chronic migraine. *Eur J Phys Rehabil Med.* 2017;53(3):342-350. PMID 28118694
27. Carvalho GF, Luedtke K, Szikszay TM, et al. Physical therapy and migraine: musculoskeletal and balance dysfunctions and their relevance for clinical practice. *Braz J Phys Ther.* 2020. PMID 31813696
28. Vives-Mestres M, Casanova A, Buse DC, et al. Patterns of Perceived Stress Throughout the Migraine Cycle: A Longitudinal Cohort Study Using Daily Prospective Diary Data. *Headache.* 2021;61(1):90-102. PMID 32918830
29. Treadwell JR, Tsou AY, Rouse B, et al. Behavioral interventions for migraine prevention: A systematic review and meta-analysis. *Headache.* 2025;65(4):668-694. PMID 39968795
30. Paudel P, Sah A. Efficacy of biofeedback for migraine: A systematic review and meta-analysis. *Complement Ther Med.* 2025;90:103153. PMID 40064357

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Educational content only. Not medical advice. Not a diagnosis. Not a substitute for your clinician. Lab interpretation belongs with someone who can see your full history.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Go to an emergency department for a headache that peaked within a minute, a headache with fever and a stiff neck, a headache with new weakness, numbness, speech trouble or vision loss, a headache after head injury, a first severe headache after age 50, or a headache in pregnancy or just after delivery. Those are not migraine behaving normally and no diet applies to them.

If you are pregnant, nursing, underweight, feeding a child, diabetic, or have a history of disordered eating, talk with a clinician before removing food groups or changing meal timing. Several of the strategies here involve restriction, and restriction is not free.

Where food sits among the five keys

Unlock Your Migraines works through five root causes: food, hormones, chronic infection, structure, and nutrients. Food is the first key, and it is the one you control three times a day, which is why it gets its own guide.

It is also the one people over-weight. If your attacks track your cycle, food will not fix that on its own. If your neck and jaw are part of the picture, no elimination diet reaches them. The lab guide maps all five keys and tells you which are yours. This guide assumes food is at least one of them.

One connection worth holding while you read: the gut and the brain talk to each other continuously, and reviews of the gut-brain axis in migraine describe that signalling as bidirectional (PMID 32054443), with microbiome-targeted approaches now under active review (PMID 41829891). That is a mechanism worth understanding, not yet a test to order or a product to buy.

What diet can and cannot do in migraine

Let me set expectations honestly, because migraine attracts more dietary folklore than almost any condition.

What diet does well. Regularity. Migraine brains dislike change, and the most reliable dietary contributions to attack frequency are not exotic foods, they are skipped meals, dehydration, inconsistent caffeine, and irregular sleep. This is unglamorous and it is where the reliable wins are.

What diet does sometimes, in some people. Removing a specific personal trigger food. This is real, and it is much more individual than the trigger lists circulating online suggest. Migraine is one of the few conditions where personalized elimination has been tested in randomized trials, and this guide covers what those trials found.

What diet does not do. Cure migraine. Migraine is substantially genetic and neurological. Diet changes how often the brain crosses its threshold, it does not change the brain.

One thing to understand before you start blaming foods. A migraine attack begins hours before the pain, in a premonitory phase that commonly includes food cravings, especially for chocolate and sweets. So the chocolate you ate the evening before an attack may have been the first symptom rather than the cause. This one insight saves people years of pointless avoidance, and it is why the reintroduction step later in this guide matters so much.

Part one: the regularity foundation

Do this part first. It is free, it has no downside, and for a meaningful share of people it is most of the available benefit.

Do not skip meals

Missed meals are among the most consistently reported migraine precipitants. A review of fasting and headache notes that while no study has tested fasting as a treatment for migraine, retrospective analyses suggest fasting may worsen migraine in the short term, and that genetic studies have found a link between glycemic processing and migraine occurrence (PMID 40117066).

Practically: eat within an hour or two of waking, do not let more than four or five hours pass in the day without food, and include protein at each meal so the interval is easier to hold. If you are attracted to intermittent fasting, understand that the current evidence on fasting in migraine points the wrong way for the short term.

Keep caffeine consistent, and stop worrying about moderate use

This is where the popular advice and the evidence part company.

A prospective cohort study followed 97 adults with physician-confirmed episodic migraine who kept twice-daily electronic diaries for six weeks. It found **no association** between habitual caffeinated beverage intake and headache frequency, duration, or intensity, and the authors state that the data do not support a recommendation that people with episodic migraine avoid caffeine (PMID 38318677).

A 2026 systematic review integrating 19 observational and Mendelian randomization studies found that lifelong genetic liability to higher coffee and caffeine intake was associated with reduced migraine risk, especially migraine with aura, while abrupt withdrawal or acute excessive intake at three or more drinks per day could trigger attacks in some people (PMID 42151857).

So the rule is consistency, not abstinence. Pick an amount, keep it the same every day including weekends, and do not swing. The Saturday morning headache after a week of workday coffee is caffeine withdrawal, and the fix is a consistent schedule rather than quitting.

Hydration and alcohol

Drink to a boringly consistent schedule rather than in reaction to thirst. Dehydration is a commonly reported precipitant and correcting it costs nothing.

Alcohol, and red wine in particular, is among the most frequently reported triggers in the migraine literature, alongside high and irregular caffeine intake (PMID 41228543). If you are running the experiments in this guide, hold alcohol steady or remove it during the trial period, otherwise it will confound everything else you are testing.

Part two: the elimination experiment, which in migraine has actual trials behind it

Personalized food elimination is contested in most conditions. In migraine it has been tested with randomized trials, and the results are the reason it is in this guide rather than in the “what I left out” section.

What the trials found

The crossover trial. Thirty patients with migraine without aura completed a randomized, double-blind, crossover, diary-based trial. After a six-week baseline, IgG antibodies against 266 food antigens were measured. Patients then spent six weeks on a diet either excluding or including their own IgG-positive foods, then crossed to the opposite diet after a two-week break. Participants averaged 24 reactive foods. During the elimination period, headache days fell from 10.5 to 7.5 and migraine attacks fell from 9.0 to 6.2 compared with baseline, both statistically significant (PMID 20647174).

The larger trial that tempers it. 167 community volunteers with migraine-like headaches were randomized to a true IgG-based elimination diet or a sham diet sheet for 12 weeks. The true diet significantly reduced headaches at four weeks. By 12 weeks the difference was no longer statistically significant, and disability and daily-life impact did not differ between the groups (PMID 21835022).

The summary in the headache literature. A narrative review of diet and headache reports that two of three randomized controlled trials found an elimination diet of IgG-positive foods significantly reduced headache or migraine frequency during treatment compared with baseline, and concludes that elimination diets can prevent headaches in subgroups of people with headache disorders (PMID 27699780).

How to read that, and how to use it

The honest read. Personalized elimination works for a subgroup, the early effect is clearer than the sustained one, and adherence is probably part of why. Nobody knows in advance which subgroup you are in, which is exactly why you run it as a bounded experiment rather than a permanent lifestyle.

The practical value of testing first. Twenty-four candidate foods out of 266 is a manageable experiment. Two hundred and sixty-six is not. Whether you use a panel or a careful diary to generate the short list, the point is the same: narrow the field, then test it properly.

What settles it is the reintroduction, not the list. Skip that step and you will end up eating twelve foods and no wiser.

The 12-week protocol

Weeks 1 and 2, baseline. Change nothing. Track four numbers daily: whether you had a headache, its severity out of ten, its duration, and whether you took acute medication. Those four are the currency of every migraine decision you will ever make.

Weeks 3 to 6, elimination. Remove your short list. Keep meals regular, caffeine constant, and sleep steady, because changing those at the same time makes the result uninterpretable. Keep tracking.

Week 7, honest review. Compare monthly headache days against your baseline. A meaningful result looks like the trial above: roughly a quarter to a third fewer headache days. If nothing moved at all, food is not your main lever. Stop the elimination, put everything back, and turn to the supplement guide and the acute medication count in the lab guide.

Weeks 8 to 12, reintroduction, one food at a time. A normal portion on one day, then three days of watching before the next. A food that reproduces the headache across two separate trials stays out. A food that does nothing goes back in permanently. Expect most to come back clean.

What you should end with: a short personal list of two or three real triggers, and your diet back. Not a spreadsheet of forbidden foods.

Part three: the specific foods worth knowing about

Aged, fermented, and histamine-rich foods

Aged cheese, cured and smoked meats, fermented foods, and alcohol are the classic list, on the reasoning that tyramine and histamine in these foods act on a system already primed to react. Histamine is broken down largely by the DAO enzyme in the gut lining, so people with limited DAO capacity handle a histamine load less well than others do.

Treat this as a well-founded hypothesis to test rather than a rule to obey. The freshness principle does most of the work: the same protein cooked and eaten today carries far less histamine than it does after three days in the refrigerator. If your headaches follow leftovers, aged cheese, and wine, this is your experiment, and DAO support is covered in the supplement guide.

Dairy

A review examining dairy and headaches found the relationship genuinely unresolved, with mechanistic plausibility through neuroinflammatory, neurotransmitter, vascular, and gut-brain pathways, and a literature limited by methodological problems and a lack of randomized trials of dairy restriction (PMID 39066996).

So: worth testing individually if your diary points there. Not worth removing on principle.

Gluten, and the celiac question that must come first

Get tested before you remove it. A systematic review and meta-analysis found an increased prevalence of celiac disease among people with idiopathic headache and vice versa, described the headaches as most often migraine-like, and found that a gluten-free diet led to total resolution of headache in up to 75% of patients in the celiac context (PMID 30301194). In a survey of 866 people with biopsy-confirmed celiac disease and headache, those with migraine improved more than those with tension-type headache after starting the diet, and compliant individuals showed a 48% improvement in headache frequency (PMID 30893404).

Celiac serology is only interpretable while you are still eating gluten. Remove it first and you have destroyed your own test. A broader narrative review notes that a gluten-free trial may help selected non-celiac patients as well (PMID 41228543), which is a fine experiment to run **after** the blood test.

Additives, and what the evidence actually supports

The narrative review above concludes that evidence for specific foods and additives remains inconsistent (PMID 41228543), and the diet and headache review found conflicting provocation-study evidence for aspartame, with two positive and two negative studies, while observational evidence gives modest support for gluten-containing and histamine-containing foods and alcohol precipitating headaches in subgroups (PMID 27699780).

Read that as permission to stop worrying about additives in general and to test the specific ones your own diary implicates.

Part four: the background pattern

Mediterranean-style eating

A narrative review synthesizing dietary interventions in migraine supports recommending Mediterranean-style eating, considering omega-3 supplementation, and using selective trials of ketogenic or elimination approaches in appropriate patients, alongside weight management and regular physical activity in people with obesity (PMID 41228543).

In practice that means vegetables and fruit at most meals, olive oil as the main fat, fish regularly, legumes and whole grains, and less ultra-processed food. It is also the pattern that makes the regularity work in Part One easier to sustain, which may be most of why it helps.

The omega-3 caveat

Fish is worth eating. Be careful about what the supplement evidence shows: in the dose-response meta-analysis of supplements for migraine prophylaxis, omega-3 supplementation **did not** produce a statistically significant reduction in any of the measured outcomes in adults (PMID 39404918). Eat the fish, and read the supplement guide before you buy the capsules.

The ketogenic option, for the right person

The review of fasting and headache reports that several randomized controlled trials of ketogenic diets significantly reduced migraine days and inflammatory markers, with improvements on disability measures, and notes that these benefits were not reproduced by simply administering ketone bodies (PMID 40117066). The dietary review likewise places ketogenic approaches among selective trials for appropriate patients (PMID 41228543).

My honest position. This is a real signal and a demanding intervention. A ketogenic diet is hard to sustain, interacts with several medications, is not appropriate in pregnancy or with certain metabolic conditions, and should be run with clinical supervision rather than from a guide. If your migraine is frequent and disabling and the simpler things have failed, it is a legitimate conversation to have. It is not where I would start.

A week that supports the plan

The rhythm matters more than the recipes: three regular meals, protein at each, nothing aged sitting in the refrigerator for days, caffeine at the same time daily.

Monday. Eggs and greens; chicken and rice bowl with olive oil; salmon, sweet potato, broccoli.

Tuesday. Oats with berries; leftovers from Monday's chicken, frozen the night before and reheated once; beef stir fry with fresh vegetables.

Wednesday. Greek-style yogurt if dairy is not on your list, or eggs if it is; lentil soup; roast chicken thighs with vegetables.

Thursday. Eggs and avocado; tuna salad with olive oil; pork chops, quinoa, green beans.

Friday. Oats with nut butter; chicken salad; fresh fish, potatoes, salad.

Saturday. Same wake time, same coffee time, even though it is the weekend. This single item prevents a common weekend attack.

Sunday. Cook ahead, freeze in single portions rather than refrigerating for the week.

What I deliberately left out, and why

- **The universal trigger list.** The lists that circulate name twenty foods for everyone. Real triggers are individual, and the premonitory craving problem means several items on those lists are consequences rather than causes.
- **Permanent elimination without reintroduction.** The failure mode of this entire category. The 12-week trial data show effect fading, which makes planned reintroduction more important, not less (PMID 21835022).
- **Detox and cleanse protocols.** No migraine outcome evidence, and fasting-adjacent approaches point the wrong way in the short term (PMID 40117066).
- **Blanket avoidance of caffeine.** Contradicted by a prospective cohort and by Mendelian randomization data (PMID 38318677, PMID 42151857). Consistency, not abstinence.
- **Blanket avoidance of aspartame and additives.** The provocation evidence is genuinely conflicting (PMID 27699780). Test yours if your diary points there.

Telling you what to skip is part of what you paid for.

When to seek professional care

Emergency: thunderclap onset, fever with stiff neck, new neurological symptoms, after head injury, first severe headache after 50, or headache in pregnancy or postpartum.

Soon:

- Acute medication on 10 or more days per month
- 4 or more migraine days per month
- Attacks that changed character or frequency
- Weight loss, or a diet that has narrowed below roughly twenty foods
- Any elimination diet in a child, before starting rather than after
- Migraine with aura while taking combined hormonal contraception

Working with Root Health

This guide is written to be safe and useful for a reader I have never met. A personalized plan starts from your diary, your labs, and your medication list, and sequences changes instead of stacking them.

Book a free Root Discovery Session: rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=migraines-diet

Consultations are available to people located in North Carolina, where Dr. Seay is licensed.

If you are outside North Carolina, the tracker is what turns this guide into an answer:

Free symptom tracker: rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=migraines-diet

Read the companion documents alongside this one. **Understanding Your Migraine Labs** covers what to test before you start eliminating, especially celiac serology, and **The Migraine Supplement Guide** covers what has been studied for prevention.

References

All references were verified against live PubMed records on 2026-08-08, including a publication type and corrections check for retractions and errata. Verification detail is documented in *Migraines-Kit-Citation-Ledger.md*.

1. Ebbert PT, Natbony LR. Fasting and Headache. *Curr Pain Headache Rep*. 2025;29(1):67. PMID 40117066
2. Mittleman MR, Mostofsky E, Vgontzas A, et al. Habitual caffeinated beverage consumption and headaches among adults with episodic migraine: A prospective cohort study. *Headache*. 2024;64(3):299-305. PMID 38318677
3. Makhoulouf HA, Khelifa H, Batarseh SF, et al. The impact of caffeine consumption on migraine: a systematic review. *BMC Neurol*. 2026;26(1). PMID 42151857
4. Poboz T, Janowski K, Michalak K, et al. Food in Migraine Management: Dietary Interventions in the Pathophysiology and Prevention of Headaches, A Narrative Review. *Nutrients*. 2025;17(21). PMID 41228543
5. Alpay K, Ertas M, Orhan EK, et al. Diet restriction in migraine, based on IgG against foods: a clinical double-blind, randomised, cross-over trial. *Cephalalgia*. 2010;30(7):829-837. PMID 20647174
6. Mitchell N, Hewitt CE, Jayakody S, et al. Randomised controlled trial of food elimination diet based on IgG antibodies for the prevention of migraine like headaches. *Nutr J*. 2011;10:85. PMID 21835022
7. Martin VT, Vij B. Diet and Headache: Part 1. *Headache*. 2016;56(9):1543-1552. PMID 27699780
8. Ceren Akgor M, Ekizoglu E, Ozge A. Dairy and Headaches: What is the Connection? *Curr Pain Headache Rep*. 2024;28(12):1255-1264. PMID 39066996
9. Zis P, Julian T, Hadjivassiliou M. Headache Associated with Coeliac Disease: A Systematic Review and Meta-Analysis. *Nutrients*. 2018;10(10):1445. PMID 30301194
10. Ameghino L, Farez MF, Wilken M, et al. Headache in Patients with Celiac Disease and Its Response to the Gluten-Free Diet. *J Oral Facial Pain Headache*. 2019;33(3):294-300. PMID 30893404
11. Talandashti MK, Shahinfar H, Delgarm P, et al. Effects of selected dietary supplements on migraine prophylaxis: A systematic review and dose-response meta-analysis of randomized controlled trials. *Neurol Sci*. 2025;46(2):651-670. PMID 39404918
12. Arzani M, Jahromi SR, Ghorbani Z, et al. Gut-brain Axis and migraine headache: a comprehensive review. *J Headache Pain*. 2020. PMID 32054443
13. Migraine and the Gut-Brain Axis, The Role of Microbiome-Targeted Biotics. *Nutrients*. 2026. PMID 41829891

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Talk with your prescriber before starting, stopping, or changing anything here. Several items in this guide have real interactions, one has a manufacturing safety requirement that can affect your liver, and one is specifically contraindicated in pregnancy. Nothing here is a reason to stop a preventive medication that is working.

Supplements in the United States are regulated as food, not as drugs. Quality varies. Look for independent third-party testing.

Why nutrient evidence looks thinner than it is

A large outcome trial costs tens to hundreds of millions of dollars, and that money only exists when a patent can pay it back. You cannot patent magnesium, an herb, or a food. So the evidence hierarchy this guide grades against is not neutral: it systematically over-documents patentable drugs and under-documents nearly everything else. That is measurable, not rhetorical. A Cochrane methodology review found that manufacturer sponsorship of drug and device studies produces more favourable results and conclusions than independent funding, through a bias that standard quality checks do not catch (PMID 28207928).

Two honest consequences follow, and they pull in opposite directions.

First, “no evidence” has three different meanings, and this guide tries to always tell you which one applies. *Tested and failed:* a real trial ran and the answer was no. *Never properly tested:* nobody could afford the trial, so absence of evidence is not evidence of absence. *Tested small and won:* real but modest trials, usually measuring markers rather than outcomes, because markers are what a small budget can afford.

Second, the asymmetry earns nutrients humility, not automatic credit. When isolated nutrients have received large publicly funded trials, the results have been mixed: some genuine wins, some nulls, and some harms. Small nutrient trials also carry their own publication bias toward positive results, just as drug trials tilt toward their sponsor. So this guide applies one rule in both directions: name the funding, name the evidence tier, and let you see both.

Two things that outrank everything in this guide

I would rather you spend your money well than spend it here, so this comes first.

One: count your acute medication days. If you take something to stop a headache on 10 or more days a month, medication overuse headache is likely part of your problem, and it will blunt everything else you try, including every supplement below. A network meta-analysis of management strategies found that abrupt withdrawal alone was **not** significantly effective, with a mean difference of -2.77 monthly headache days and a confidence interval crossing zero, while withdrawal paired with a preventive strategy was effective, and combination approaches including anti-CGRP therapies and nerve blocks ranked highest (PMID 40011869). Note the shape of that: quitting cold is not the answer, quitting with a plan is.

Two: if you have four or more migraine days a month, know that modern preventives exist. A network meta-analysis of 16 randomized trials covering 9,123 participants found CGRP monoclonal antibodies effective for preventing episodic migraine, with meaningful reductions in monthly migraine days and differing efficacy and safety profiles between agents (PMID 41546694). These are a different order of intervention from anything in this guide.

Supplements in migraine are genuinely useful, and they are an addition to that foundation rather than a substitute for it. Several of the items below have decades of use behind them and are reasonable first steps for someone with a handful of attacks a month who does not want a daily prescription. None of them will match a modern preventive in someone with frequent, disabling migraine.

How to read this guide

The tiers. Tier 1 has the most consistent randomized evidence in migraine prevention. Tier 2 has real trials with meaningful limitations. Tier 3 needs a prescriber conversation first.

Typical studied range is the dose used in the cited research, given so you can discuss it intelligently. It is not a prescription for you.

Give it three months. This is the most-broken rule in migraine supplementation. Nearly every trial below ran for at least eight to twelve weeks, and riboflavin in particular is known for taking months. People quit at week three and conclude it failed.

Track four numbers daily: headache yes or no, severity out of ten, duration in hours, and whether you took acute medication. Monthly migraine days is the outcome every trial in this guide measured, and it should be the one you measure too.

One change at a time. Migraine frequency varies enormously month to month on its own. If you start three things at once you will never know which one earned its place.

Foundation: the two preventives that are not products

These come first because they have systematic review evidence behind them, they cost little, and they act on drivers rather than on symptoms.

Behavioral prevention, which is a real preventive

A systematic review and meta-analysis of behavioral interventions for migraine prevention synthesized the benefits and harms in children and adults, informed by an expert panel including neurologists, psychologists, and people living with migraine (PMID 39968795). A separate systematic review and meta-analysis assessed biofeedback specifically, in which patients learn to modify physiological parameters through feedback-mediated awareness, promoting relaxation and reducing stress (PMID 40064357).

This matters because stress is not a bystander in migraine. A longitudinal cohort using daily prospective diaries described how perceived stress patterns move through the stages of the migraine cycle (PMID 32918830).

What to actually do. Pick one and do it daily for three months, the same window you would give a supplement: a relaxation practice, a paced breathing practice, or biofeedback if you have access to it. Track it in the same diary as your headache days so you can see whether it earned its place.

The structural half, which is my own field

Neck pain is prevalent in migraine, and musculoskeletal dysfunctions are prevalent enough to be a legitimate treatment target rather than an incidental finding (PMID 37331926). Women with episodic and chronic migraine differ from controls on measures of upper cervical mobility, symptomatic upper cervical joints, and head and neck posture (PMID 28118694), and reviews of physical therapy in migraine describe both musculoskeletal and balance dysfunction as clinically relevant (PMID 31813696).

What this is and is not. It is a reason to have your neck, jaw, and posture assessed as part of a migraine workup, especially with neck pain, clenching, or a long desk posture. It is not a claim that an adjustment cures migraine, and neck involvement in migraine is often as much consequence as cause.

Tier 1: the best-supported three

Magnesium

What it is. A mineral involved in nerve signaling, vascular tone, and cortical excitability. Common forms include glycinate, citrate, malate, and oxide, which differ mainly in absorption and how likely they are to loosen your stool.

What the research shows. A systematic review and dose-response meta-analysis of 22 randomized controlled trials of supplements for migraine prophylaxis found that magnesium reduced migraine attacks with a mean difference of -2.51, reduced severity by -0.88, and reduced monthly migraine days (PMID 39404918). An earlier meta-analysis found that oral magnesium significantly alleviated the frequency and intensity of migraine, and separately that intravenous magnesium relieved acute attacks at 15 to 45 minutes, 120 minutes, and 24 hours after infusion (PMID 26752497).

The honest counterweight. A systematic review and meta-analysis in a neurology journal reached a more cautious conclusion, finding that on the available data it is unknown whether coenzyme Q10 and magnesium are effective for migraine prophylaxis in adults, and specifically that magnesium did not significantly decrease migraine severity across three trials in 226 participants (PMID 30764890).

So: the most recent and largest synthesis is positive, an earlier careful one is agnostic, the risk is low and the cost is low. That combination is why magnesium is still where I start.

Typical studied range. Trials commonly use 400 to 600 mg of elemental magnesium daily, and the dose-response analysis above examined the dose relationship directly (PMID 39404918).

Timing. Evening suits most people. Split the dose if your gut objects.

Cautions. Loose stools are the usual limiting effect, and citrate is the worst offender while glycinate is generally the gentlest. Magnesium interferes with absorption of some antibiotics, thyroid medication, and bisphosphonates, so separate them by several hours. **If you have reduced kidney function, do not take magnesium without your clinician's approval.**

Riboflavin (vitamin B2)

What it is. A B vitamin your mitochondria use in energy production. The migraine rationale is mitochondrial: if the migraine brain has an energy-handling problem, riboflavin supports the machinery.

What the research shows. A systematic review and meta-analysis of nine studies, eight of them randomized controlled trials, covering 673 subjects, found that riboflavin supplementation significantly decreased migraine days, attack duration, attack frequency, and pain score. The pooled analysis specifically supports 400 mg per day for three months (PMID 33779525).

Typical studied range. 400 mg daily for three months (PMID 33779525). That is a high dose relative to a multivitamin, and it is the dose that was studied.

Timing. With food, and in the morning, since some people find it mildly activating.

Cautions. It turns your urine bright yellow. That is riboflavin, it is harmless, and it is worth knowing before it alarms you. Riboflavin is water-soluble and well tolerated, and it is one of the few things in this guide with almost no interaction profile. Ask first in pregnancy, as with everything.

Coenzyme Q10

What it is. A compound in the mitochondrial electron transport chain. Same mitochondrial rationale as riboflavin.

What the research shows. A meta-analysis of six studies with 371 participants found that CoQ10 reduced the duration of headache attacks and reduced the frequency of migraine attacks with a mean difference of -1.52 attacks, while finding **no statistically significant reduction in severity** (PMID 33402403). The dose-response analysis of supplements likewise included CoQ10 among the agents with measurable prophylactic effect (PMID 39404918), and the more cautious neurology review concluded that its effectiveness is unknown on the available evidence (PMID 30764890).

How to read it. Fewer and shorter attacks, not less painful ones. That is a real and useful benefit, and it is a narrower claim than the marketing makes.

Typical studied range. Trials commonly use 300 mg daily, often in divided doses.

Timing. With a meal containing fat, since absorption depends on it. Earlier in the day, as some find it activating.

Cautions. CoQ10 can reduce the effect of warfarin, which requires prescriber involvement. It may modestly lower blood pressure and blood sugar. Not established in pregnancy or nursing.

Tier 2: real trials, real limitations

Melatonin

What the research shows. A systematic review and meta-analysis of nine randomized controlled trials in 788 patients found that melatonin reduced attack duration compared with placebo, by a mean of about 5 hours, along with improvements in patient-reported outcomes, and a favourable safety profile. Compared with amitriptyline, melatonin was generally less effective for attack duration and severity, but had better tolerability including a lower risk of sleepiness (PMID 41627537). An earlier systematic review and meta-analysis likewise concluded that melatonin may be of potential benefit in migraine prevention in adults while calling for higher-quality trials, and noted a superior safety profile compared with amitriptyline in both adults and children (PMID 32352572).

Why I like it in the right person. If your attacks cluster around poor or irregular sleep, this addresses two problems at once. The trials most often used 3 mg at night.

Cautions. Next-morning grogginess usually means the dose is too high. It stacks with other sedating medication. Not established as safe in pregnancy or nursing.

Feverfew

What the research shows. A Cochrane review of feverfew for preventing migraine found that a larger rigorous study added since the previous version reported a difference of 0.6 attacks per month between feverfew and placebo, describing this as low quality evidence that adds some positive weight to previously mixed and inconclusive findings, and needing confirmation in larger trials with stable extracts. It also found feverfew was not associated with any major safety concerns, with only mild and transient adverse events, most commonly gastrointestinal complaints and mouth ulcers (PMID 25892430).

How to read it. Half an attack a month, on low quality evidence. Genuinely modest. It is on this list because it is well tolerated and because the honest number is the useful thing to know before you buy it.

Cautions. Feverfew is in the daisy family, so people allergic to ragweed, chrysanthemum, marigold, or daisy may react. It may affect bleeding, so it matters with anticoagulants and before surgery. **Avoid in pregnancy.** Stopping abruptly after long use has been associated with a rebound syndrome, so taper.

Butterbur (*Petasites hybridus*)

What the research shows. A three-arm randomized trial compared Petasites extract 75 mg twice daily, 50 mg twice daily, and placebo in 245 patients with migraine over four months. Attack frequency fell 48% with the 75 mg dose against 26% for placebo, and 68% of patients on 75 mg achieved at least a 50% reduction in attack frequency versus 49% on placebo. The 50 mg dose was not significantly better than placebo on the primary endpoints. The most common adverse effects were mild gastrointestinal events, predominantly burping (PMID 15623680).

That is one of the strongest results for any natural product in migraine prevention. It comes with a serious condition attached.

Cautions, and this one is not optional. Raw butterbur contains pyrrolizidine alkaloids, which are liver toxins. Only preparations processed to remove them are appropriate for human use, and that is a manufacturing question you cannot answer by looking at the bottle. **If you cannot confirm from the manufacturer that the product is certified free of pyrrolizidine alkaloids, do not take it.** Butterbur is also in the daisy family, so the ragweed cross-reactivity above applies. Not for use in pregnancy or nursing. Discuss it with your clinician if you have any liver condition, and stop it and call if you develop yellowing of the skin or eyes, dark urine, or right-sided abdominal pain.

Vitamin D, with a probiotic

What the research shows. In the dose-response meta-analysis, vitamin D reduced migraine frequency by a mean of -1.69 attacks and monthly migraine days by -2.41, and probiotics reduced frequency by -1.16, severity by -1.07, and monthly migraine days by -3.02 (PMID 39404918).

Two randomized, triple-blinded, placebo-controlled trials tested the combination directly in adults with migraine diagnosed by international criteria. Twelve weeks of vitamin D at 50,000 IU every two weeks plus a multispecies probiotic produced significantly greater reductions in headache frequency and severity than placebo, while headache duration, hs-CRP, and mood and impact questionnaire scores did not differ (PMID 39394141). The companion trial found a significantly greater reduction in a composite migraine index in the supplemented group (PMID 40360751).

Typical studied range. As above, with the caveat that 50,000 IU every two weeks is a clinician-directed dose, not a self-selected one. Test your level first, and read the lab guide.

Cautions. Vitamin D is fat-soluble and accumulates. It interacts with thiazide diuretics, some seizure medications, and digoxin. Probiotics are generally well tolerated, but people who are significantly immunocompromised or critically ill should not start them without medical guidance.

B vitamins, specifically for migraine with aura

What the research shows. A randomized, double-blind, placebo-controlled trial gave 52 patients with migraine with aura folic acid 2 mg, vitamin B6 25 mg, and vitamin B12 400 micrograms daily for six months. Homocysteine fell 39%, significantly more than placebo, and headache frequency and pain severity were reduced in the treated group while the placebo group did not change. Response tracked MTHFR C677T genotype, with C allele carriers responding better than TT genotypes (PMID 19384265). A larger 206-patient trial of the same combination found that MTHFR C677T and MTRR A66G variants independently influenced response (PMID 22926161). In a cohort of 141 women with migraine with aura, higher dietary folate intake was inversely associated with migraine frequency (PMID 25598270).

Who this is for. Migraine **with aura**, particularly with an elevated homocysteine. That is a narrower population than most supplement marketing implies, and within it this is some of the better-targeted evidence in the whole guide.

Cautions. Long-term high-dose B6 can cause peripheral nerve damage, which is the one genuinely important caution here. The trial dose above was 25 mg daily, well below the doses associated with neuropathy, and there is no reason to exceed it. If you develop numbness or tingling in the hands or feet, stop and call.

Ginger, for the attack rather than for prevention

What the research shows. A double-blind randomized clinical trial in 100 patients with acute migraine without aura compared ginger powder against sumatriptan across five attacks. Two hours after either treatment, mean headache severity decreased significantly, efficacy was statistically comparable between the two, adverse effects were fewer with ginger, and patient satisfaction and willingness to continue did not differ (PMID 23657930).

How to read it. One trial, 100 patients, and a genuinely interesting result. It is worth having in the house for a mild attack, especially if you are trying to reduce acute medication days for the reasons at the top of this guide. It is not a replacement for a triptan in a severe attack, and this is one study rather than a literature.

Cautions. Ginger can affect bleeding at higher doses, which matters with anticoagulants and before surgery. It commonly causes heartburn.

Tier 3: ask your prescriber first

- **Butterbur**, for the pyrrolizidine alkaloid issue. The only item here where a manufacturing failure can injure your liver.
- **Feverfew in pregnancy**, where it should be avoided outright.
- **High-dose vitamin D** without a blood level.
- **Anything in the daisy family**, including feverfew, butterbur, and chamomile, if you are ragweed allergic.
- **Anything at all in pregnancy or nursing.** Migraine often changes in pregnancy and the management is genuinely different. That is an obstetric conversation.
- **Any change to a preventive medication.** Nothing here is a reason to reduce one on your own.

What I deliberately left out, and why

- **Omega-3 supplements for migraine prevention.** In the dose-response meta-analysis, omega-3 supplementation did not produce a statistically significant reduction in any measured outcome in adults (PMID 39404918). Eat fish for other reasons.
- **Combination “migraine formulas” at token doses.** The doses that worked in trials are specific: 400 mg riboflavin, 300 mg CoQ10, 400 to 600 mg magnesium. Multi-ingredient blends routinely contain a fraction of each and cost more than buying the two or three that are actually supported.
- **Detox protocols and IV drips marketed for migraine.** No prophylaxis evidence, real cost. Intravenous magnesium is a separate matter with actual acute-attack data behind it (PMID 26752497), and it belongs in a medical setting rather than a wellness clinic drip menu.
- **Stopping caffeine as a supplement strategy.** Covered in the diet guide: consistency beats abstinence, and the prospective data do not support blanket avoidance.
- **Starting five products at once.** Migraine frequency swings on its own. A stack guarantees you learn nothing.

Telling you what to skip is part of what you paid for.

Running a fair three-month trial

Weeks 1 to 4, baseline. Change nothing. Track the four daily numbers. Count acute medication days, because that number decides whether anything else in this guide can work.

Weeks 5 to 16, one intervention. Start with magnesium if you are unsure, riboflavin if attacks are frequent and you can commit to three months, and CoQ10 if fatigue travels with your attacks. Low end of the range first.

If you menstruate, judge against the cycle, not the calendar month. Menstrual migraine is attributed to premenstrual estrogen withdrawal (PMID 37730536) and is recognized as a distinct disorder (PMID 33600767). A supplement trial that happens to span two heavy cycle months will look like a failure that was never one. Map attacks against cycle days, and compare like with like.

Week 17, judge it. Compare monthly migraine days against your baseline month. A 30% reduction is a real result. Anything under about 10% is noise.

Then, and only then, add a second. Same rules.

Bring the diary to every appointment. Three months of daily numbers is the most useful thing you can hand a clinician, and it is exactly what a preventive medication decision is based on.

Realistic expectations

What is realistic. A few fewer migraine days a month. Shorter attacks. Less acute medication. In the trials above, the effects ranged from about half an attack per month for feverfew up to two or three fewer monthly migraine days for the better-supported items. Those are worth having, and they stack with sleep, meal regularity, and consistent caffeine.

What is not realistic. Elimination of migraine. Nothing in this guide does that, and neither does any prescription.

Timeline. Three months. Riboflavin especially.

Cost sense. If something has done nothing after three months at the studied dose, stop paying for it. That money is better spent on the preventive conversation, which is where the larger effects live.

When to seek professional care

Emergency: thunderclap onset, fever with stiff neck, new neurological symptoms, after head injury, first severe headache after 50, headache clearly triggered by lying flat or standing, or headache in pregnancy or postpartum.

Soon:

- Acute medication on 10 or more days per month
- 4 or more migraine days per month
- Attacks that changed in character, frequency, or severity
- Aura lasting more than an hour, or new aura after age 50
- Migraine with aura while taking combined hormonal contraception

Stop and call your prescriber for: yellowing of the skin or eyes, dark urine, or right-sided abdominal pain, which are liver warning signs and matter most with butterbur. Also for new numbness or tingling in the hands or feet while taking B6, unusual bruising or bleeding, or a new rash.

Where to find quality versions of these

Nothing in this guide names a brand, on purpose. The evidence supports ingredients and doses, not labels. Here is what to look for.

Look for third-party testing. NSF, USP Verified, or an equivalent independent seal means someone outside the company confirmed the bottle contains what the label claims. This matters most for butterbur, where the pyrrolizidine question is a safety question rather than a quality preference.

Check the dose against the trial, not against the marketing. 400 mg riboflavin, 300 mg CoQ10, 400 to 600 mg elemental magnesium. Many products carry a fraction of these.

Check the form. “Magnesium” on the front can be oxide, which is mostly a laxative, or glycinate, which is the one most people tolerate.

If you would rather not evaluate all of that yourself, Root Health keeps a curated dispensary of professional-grade products, screened for the standards above and organized by the categories in this guide.

Root Health dispensary: us.fullscript.com/welcome/drseay

Disclosure: Root Health earns a portion of dispensary purchases. That revenue supports the practice, and it is also why this guide was written to be complete and usable on its own. Every recommendation here is based on the published evidence cited in these pages, and nothing was included, excluded, ranked, or dosed based on what it earns. You can buy any of these anywhere, and the two highest-value actions in this guide, cutting acute medication days and keeping a diary, earn us nothing at all.

Working with Root Health

Everything here is general education, written to be safe for a reader I have never examined. A personalized plan starts from your diary, your labs, and your full medication list, and sequences interventions rather than stacking them.

***Book a free Root Discovery Session: [rootcauseunlocked.com/discovery?](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=migraines-supplement)
[utm_source=guide&utm_medium=ebook&utm_campaign=migraines-supplement](https://rootcauseunlocked.com/discovery?utm_source=guide&utm_medium=ebook&utm_campaign=migraines-supplement)***

Consultations are available to people located in North Carolina, where Dr. Seay is licensed.

If you are outside North Carolina, track first. Three months of daily numbers is what turns “I think it helped” into a decision:

***Free symptom tracker: [rootcauseunlocked.com/tracker?](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=migraines-supplement)
[utm_source=guide&utm_medium=ebook&utm_campaign=migraines-supplement](https://rootcauseunlocked.com/tracker?utm_source=guide&utm_medium=ebook&utm_campaign=migraines-supplement)***

Read the companion documents alongside this one. **Understanding Your Migraine Labs** covers what to test before you supplement, and **The Migraine Diet Guide** covers meal timing, caffeine, and the elimination trial.

References

All references were verified against live PubMed records on 2026-08-08, including a publication type and corrections check for retractions and errata. Verification detail is documented in `Migraines-Kit-Citation-Ledger.md`.

1. Koonalintip P, Yamutai S, Setthawatcharawanich S, et al. Network meta-analysis comparing efficacy of different strategies on medication-overuse headache. *J Headache Pain*. 2025;26(1):43. PMID 40011869
2. Shakir I, Shahzad F, Shabbir A, et al. Efficacy and safety of CGRP monoclonal antibodies for migraine prevention in episodic migraine; a network Meta-analysis. *Eur J Clin Pharmacol*. 2026;82(2):27. PMID 41546694
3. Talandashti MK, Shahinfar H, Delgarm P, et al. Effects of selected dietary supplements on migraine prophylaxis: A systematic review and dose-response meta-analysis of randomized controlled trials. *Neurol Sci*. 2025;46(2):651-670. PMID 39404918
4. Chiu HY, Yeh TH, Huang YC, et al. Effects of Intravenous and Oral Magnesium on Reducing Migraine: A Meta-analysis of Randomized Controlled Trials. *Pain Physician*. 2016;19(1):E97-112. PMID 26752497
5. Okoli GN, Rabbani R, Kashani HH, et al. Vitamins and Minerals for Migraine Prophylaxis: A Systematic Review and Meta-analysis. *Can J Neurol Sci*. 2019;46(2):224-233. PMID 30764890
6. Chen YS, Lee HF, Tsai CH, et al. Effect of Vitamin B2 supplementation on migraine prophylaxis: a systematic review and meta-analysis. *Nutr Neurosci*. 2022;25(9):1801-1812. PMID 33779525
7. Sazali S, Badrin S, Norhayati MN, et al. Coenzyme Q10 supplementation for prophylaxis in adult patients with migraine, a meta-analysis. *BMJ Open*. 2021;11(1):e039358. PMID 33402403
8. Abouelmagd ME, Aldemerdash MA, Khatatbeh AA, et al. Efficacy and Safety of Melatonin in Migraine Prophylaxis: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Curr Pain Headache Rep*. 2026;30(1):25. PMID 41627537
9. Liampas I, Siokas V, Brotis A, et al. Endogenous Melatonin Levels and Therapeutic Use of Exogenous Melatonin in Migraine: Systematic Review and Meta-Analysis. *Headache*. 2020;60(7):1273-1299. PMID 32352572
10. Wider B, Pittler MH, Ernst E. Feverfew for preventing migraine. *Cochrane Database Syst Rev*. 2015;4(4):CD002286. PMID 25892430
11. Lipton RB, Gobel H, Einhaupl KM, et al. Petasites hybridus root (butterbur) is an effective preventive treatment for migraine. *Neurology*. 2004;63(12):2240-2244. PMID 15623680
12. Tirani SA, Khorvash F, Saneei P, et al. Effects of probiotic and vitamin D co-supplementation on clinical symptoms, mental health, and inflammation in adult patients with migraine headache: a randomized, triple-blinded, placebo-controlled trial. *BMC Med*. 2024;22(1):457. PMID 39394141
13. Tirani SA, Saneei P, Khorvash F, et al. Effects of probiotic and vitamin D co-supplementation on migraine index, quality of life, and oxidative stress in adults with migraine headache: a randomized triple-blinded clinical trial. *Eur J Nutr*. 2025;64(4):179. PMID 40360751
14. Lea R, Colson N, Quinlan S, et al. The effects of vitamin supplementation and MTHFR (C677T) genotype on homocysteine-lowering and migraine disability. *Pharmacogenet Genomics*. 2009;19(6):422-428. PMID 19384265
15. Menon S, Lea RA, Roy B, et al. Genotypes of the MTHFR C677T and MTRR A66G genes act independently to reduce migraine disability in response to vitamin supplementation. *Pharmacogenet Genomics*. 2012;22(10):741-749. PMID 22926161

16. Menon S, Lea RA, Ingle S, et al. Effects of dietary folate intake on migraine disability and frequency. *Headache*. 2015;55(2):301-309. PMID 25598270
17. Maghbooli M, Golipour F, Moghimi Esfandabadi A, et al. Comparison between the efficacy of ginger and sumatriptan in the ablative treatment of the common migraine. *Phytother Res*. 2014;28(3):412-415. PMID 23657930
18. Treadwell JR, Tsou AY, Rouse B, et al. Behavioral interventions for migraine prevention: A systematic review and meta-analysis. *Headache*. 2025;65(4):668-694. PMID 39968795
19. Paudel P, Sah A. Efficacy of biofeedback for migraine: A systematic review and meta-analysis. *Complement Ther Med*. 2025;90:103153. PMID 40064357
20. Vives-Mestres M, Casanova A, Buse DC, et al. Patterns of Perceived Stress Throughout the Migraine Cycle: A Longitudinal Cohort Study Using Daily Prospective Diary Data. *Headache*. 2021;61(1):90-102. PMID 32918830
21. Luedtke K, Carvalho G, Szikszay T. Musculoskeletal dysfunctions and physiotherapy treatment strategies in patients with migraine. *Musculoskelet Sci Pract*. 2023;66:102805. PMID 37331926
22. Ferracini GN, Florencio LL, Dach F, et al. Musculoskeletal disorders of the upper cervical spine in women with episodic or chronic migraine. *Eur J Phys Rehabil Med*. 2017;53(3):342-350. PMID 28118694
23. Carvalho GF, Luedtke K, Szikszay TM, et al. Physical therapy and migraine: musculoskeletal and balance dysfunctions and their relevance for clinical practice. *Braz J Phys Ther*. 2020. PMID 31813696
24. Raffaelli B, Do TP, Chaudhry BA, et al. Menstrual migraine is caused by estrogen withdrawal: revisiting the evidence. *J Headache Pain*. 2023;24(1):131. PMID 37730536
25. Ornello R, Caponnetto V, Ratti S, et al. Menstrual migraine: a distinct disorder needing greater recognition. *Lancet Neurol*. 2021. PMID 33600767
26. Lundh A, Lexchin J, Mintzes B, et al. Industry sponsorship and research outcome. *Cochrane Database Syst Rev*. 2017;2(2):MR000033. PMID 28207928

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